Assignment 3: From Pthreads to OpenMP :)

Fall 2026

Due: Monday, Nov 9, 11:59 pm

For the following use your previous sequential submission from Assignment 1 as your starter code. Make sure that your sequential version works as expected first!

Coding Task 1: Convert the Array Sum program to parallel using OpenMP.

Coding Task 2: Convert the Matrix Multiplication program to parallel using OpenMP.

Coding Task 3: Convert the numerical integration using trapezoidal rule to parallel using OpenMP.

Coding Task 4: Convert the N Queens program to parallel using OpenMP.

Task 5: Benchmark the time it took for your sequential, Pthread, and OpenMP versions for tasks 1, 2, 3 and 4 using the table below:

NOTE: Please use **time ./a.out** or **Clock (MONOTONIC, ..)** for timing.

For OpenMP use **omp_get_wtime()** as follows:

```c
// Start timing
double start_time = omp_get_wtime();

// Some parallel region
#pragma omp parallel
{
    // parallel stuff
}

// End timing
double end_time = omp_get_wtime();

printf("Time taken: %f seconds\n", end_time - start_time);
```

Array Sum (for N 1,000,000,000)						
	Sequential	1 thread	2 threads	5 threads	7 threads	9 threads
Time						
pthread						
OpenMP						
Matrix Multiplication (for matrix size 1,000)						
	Sequential	1 thread	2 threads	5 threads	7 threads	9 threads
Time						
pthread						
OpenMP						
Trapezoidal Rule Integration (for interval size 1,000,000,000)						
	Sequential	1 thread	2 threads	5 threads	7 threads	9 threads
Time						
pthread						
OpenMP						
N Queens (for N = 15)						
	Sequential	1 thread	2 threads	5 threads	7 threads	9 threads
Time						
pthread						
OpenMP						

[D2L Discussion Post] Task 6: Explain the following:

- The differences you observed between pthreads and OpenMP in terms of control, complexity, and performance timing.
- Experiment with static, dynamic, and guided scheduling and report on the impact on performance.

Deliverables:

- A zipped folder named A4 submitted to D2L dropbox that contains:
    - Your source code files for all coding tasks.
    - A pdf file for screenshots showing your codes work as expected and for task 5.
    - A discussion post for Task 6 under D2L discuss – OpenMP.
    - A README.txt file
- Available on GitHub inside the folder where you added me as a collaborator.

Total Points (100)

- Code runs and works as expected – Task 1: 20 points
- Code runs and works as expected – Task 2: 20 points
- Code runs and works as expected – Task 3: 20 points
- Code runs and works as expected – Task 4: 20 points
- Clear documentation – Task 5: 5 points
- Clear, detailed explanation & replied to at least 2 peer posts – Task 6: 10 points
- Proper commenting and screenshot of code output: 3 points
- Available on GitHub: 2 points